

pHigment: Designing Biodegradable Single-Use Cosmetics for On-body Chemical Sensing

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Figure 1: pHigment as a sustainable Biocosmetic Interface: (a) pH values and prototypes: eyeliner, nails, and tattoos; (b) mobile app displaying a pH reading; (c) butterfly tattoo with pH 4.5 (pink hue, left) and pH 9.0 (blue hue, right); (d) nail polish sticker with different pH levels; (e) eyeliner stickers with top unreacted and bottom exposed to pH 9.0; (f) composting sample.

Abstract

pHigment explores the dual design challenge of creating on-body interfaces that are both environmentally sustainable and capable of on-body chemical sensing. We present a set of single-use cosmetic prototypes—eyeliner, temporary tattoos, and press-on nails—that respond to chemical changes in body and external fluids while being compostable after use. Each prototype is fabricated with skin-safe materials such as using alginate as a biodegradable substrate, and red cabbage powder as the anthocyanin colorimetric biosensor. This project identifies 10 expected affordances that support its use as a single-use biosensing interface. We describe the fabrication process and introduce a system that includes a custom-designed case and a computer vision-based mobile application. The app interprets color changes and outputs corresponding pH values. We report technical evaluations of the materials, system functionality, and compostability. This work contributes to the design of biocosmetic interfaces by demonstrating how ephemeral body-worn form factors can simultaneously serve as both responsive chemical sensors and components of a circular materials strategy.

CCS Concepts

• **Human-centered computing** → Interactive systems and tools; Ubiquitous and mobile computing; • **Applied computing** → Life and medical sciences.

Keywords

biocosmetic interfaces, pH sensing, biodegradable materials, colorimetric biosensors, sustainable HCI, wearable devices

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1 Introduction

pHigment proposes sustainable and chemical-based sensing through single-use cosmetics by integrating biosensors and biomaterials into temporary tattoos, eyeliner stickers, and press-on nails. These form factors often contact body fluids, enabling diverse body placement, customizable shapes, and color ranges that support visual transitions of colorimetric biosensors. Unlike conventional cosmetics that absorb into the skin or wear off, these products retain



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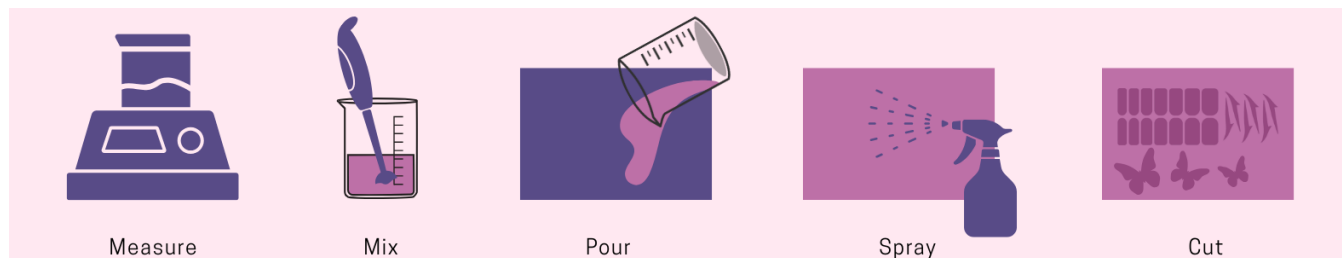


Figure 2: Fabrication process of pHigment material: measure ingredients, blend with an immersion blender, cast and level onto acrylic sheets, crosslink with calcium chloride spray, and cut wearable shapes using a vinyl cutter.

their material structure and are typically discarded after brief use, contributing to landfill waste due to synthetic components.

Colorimetric biosensors have been widely used in medical and HCI research to visually indicate the presence of specific analytes [10, 15, 30]. Examples include bandages or temporary tattoos that change color in response to pH, glucose, or lactate. In HCI, these sensors have been embedded into permanent tattoos [28], lipsticks [18], dental braces [23], and temporary tattoos [12, 16] to support integration with daily routines. These interfaces support biofeedback, and aligns with aesthetic and social practices. Recent work emphasizes the importance of biosensing systems that are not only functional but also materially and socially integrated into everyday life [31]. Our project builds on this work using a natural, skin-safe, food-grade pH indicator derived from red cabbage powder.

Sustainability has also become an increasing concern in HCI, particularly around biomaterials. Prior work presented biodegradable substrates and responsive biomaterials such as bio-dissolving fibers [11, 31]. MycoAccessories embedded electronics into mycelium [22]. BioENails explored the sustainable artificial nails using agar and chitosan with embedded NFC chips and thermochromic pigments [21]. Alganyl introduced biodegradable garments using alginate-based films with UV, thermochromic and conductive properties [3]. Holo-Chemie used alginate sheets with biosensors coated on the material surface [16]. In contrast, our biosensor is embedded within the material for improved color stability, avoiding additional coatings or user-applied layers. While prior HCI work emphasized biodegradable substrates for electronics or thermochromic responses, our focus is on biomaterials for single-use cosmetics with fluid-based biosensing capabilities. In doing so, this work aligns with the Triple Bottom Line framework [17] by addressing environmental impact through compostable materials (planet), enabling health-related self-monitoring (people), and proposing a novel category of sustainable biocosmetics interfaces with potential for future innovation (prosperity).

Beauty Technology [27] first integrated electronics into cosmetics on nails[26], hair[25], skin[8, 24]. Biocosmetic Interfaces [18, 19] shifted toward biosensing with cosmetics like lipstick, nail polish, tattoos, and dental braces for discreet health monitoring. This work expands this concept with biomaterials through four main contributions:

(1) The biosensor-biomaterial duality for biocosmetic interfaces: Novel form factors—eyeliner, temporary tattoos, and press-on nails—

demonstrate how ephemeral cosmetic products can serve as both responsive biosensors and compostable, body-worn devices;

(2) A DIY, skin-safe, biodegradable fabrication process based on alginate formulations with embedded anthocyanin;

(3) A detection system combining a custom case for light control and a computer vision app that converts RGB color to pH value;

(4) A compostability test of the material under natural conditions.

2 Implementation

2.1 Affordances for Sustainable Biocosmetic Interfaces

We approached the design of pHigment by identifying 10 expected affordances that support its use as a single-use biosensing interface:

Body-fluid interaction: We selected three form factors based on their common use on different parts of the body and their exposure to fluids: temporary tattoos, eyeliner stickers, and press-on nails. Tattoos support sweat detection across the body; eyeliner interacts with sweat and tears; nails engage with various fluids, including saliva, beverages, and environmental residues.

Colorimetric visibility: Anthocyanin provides visible color transitions, hues match the typical color palettes in cosmetics, and physiologically relevant pH range (Fig. 1.a).

Material safety and accessibility: The materials are accessible, non-toxic, skin-compatible, and safe for external body contact.

Biodegradable Use: Sodium alginate serves as the structural base for its known biodegradability under natural conditions, with visible degradation observed in two weeks.

Flexible Fit: The alginate mixture produces flexible, conformable films that can mimic the texture and behavior of conventional single-use cosmetics.

DIY Customization: It was designed for replication using household tools. Sheets can be fabricated with different thicknesses and cut into various shapes.

Wear Duration: Usually are short-term and single-use: eyeliner 1 day, nail stickers 3-5 days, tattoo stickers 2-5 days.

Reliable Sensing: Anthocyanin in the alginate matrix improves color stability and prevents degradation or detachment.

Lighting control: An enclosure ensures proper camera alignment and shields from ambient light for accurate sensing.

Digital Interpretation: A smartphone app converts RGB values to estimated pH with printed color references for calibration.

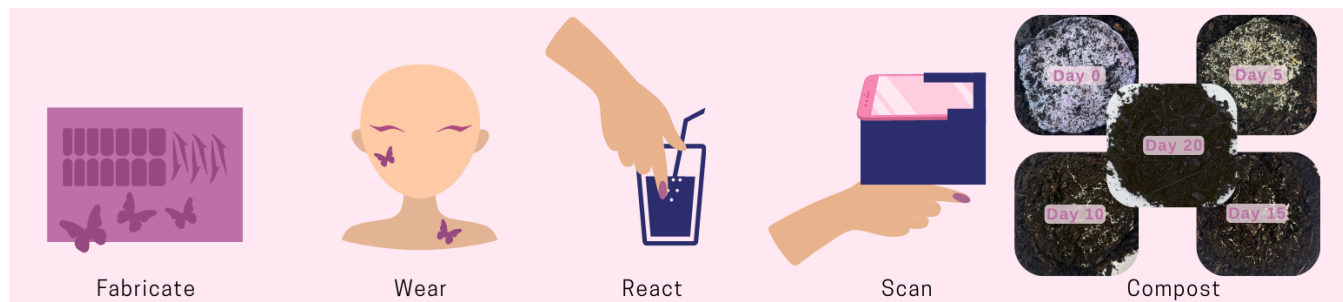


Figure 3: Material lifecycle of pHigment: fabricated material, applied to the body as a sticker, exposed to liquid for sensing, scanned using a smartphone and light box, and ultimately disposed into compost where it biodegrades.

2.2 pHigment Fabrication Process

We created flexible, pH-responsive sheets by embedding anthocyanin directly into a biodegradable alginate matrix. The formulation materials and their functions are shown in Table 1. The mixture was blended with an immersion blender, degassed using a vacuum chamber, and cast onto acrylic sheets. After leveling, a calcium chloride solution was sprayed to crosslink the alginate. Sheets were left to dry at room temperature for six days. Alternatively, a dehydrator can be used to accelerate drying; we found that drying at 100°F (approximately 38°C) for 3 hours also produced similar results, though this step is not strictly necessary. Once dried, the material was mounted onto sticker backing paper and cut into eyeliner, tattoo, and nail shapes using a vinyl cutter. These forms can be applied directly to the body for pH sensing.

Table 1: Bioplastic substrate formulation

Ingredient	Amount	Function
Sodium Alginate	3g	Polymer
Distilled Water	100g	Solvent
Glycerin	5g	Plasticizer
Sunflower Oil	2.5g	Filler
Red Cabbage Powder	0.25g	pH-reactive dye
Titanium Dioxide	0.1g	Opacifier
Calcium Chloride	10% w/v	Cross-linking agent

2.3 Biosensor Detection System

The pHigment system¹ uses a smartphone to capture images and estimate pH levels from the color-changing cosmetic samples. It also includes a custom 3D-printed light box with two white LEDs, a printed color reference chart, and a hole for placing the sample.

Smartphone cameras are a low-cost option for detecting color changes compared with spectrophotometers, but results can vary depending on lighting and camera settings [20]. The light box is based on a previous system that used a similar setup and internal color calibration chart[9]. We manually set camera exposure to avoid color shifts caused by auto settings, as recommended by

[20]. Our app includes a tap-to-sample interaction and uses image segmentation methods inspired by [5].

To use the system, the user places the cosmetic sample under the hole of the light box and takes a photo. The image includes six colors from the reference patches (white, gray, red, green, blue, and yellow) and the sample. The reference patches are measured using a lab-graded spectrophotometer to provide known RGB values. The app first applies white balance and color correction using these patches. Then, it extracts RGB values from the sample area and compares them to the closest value in an RGB table of known pH levels. These values were created using calibration solutions measured with a pH meter and photographed under the same lighting box. The app displays the estimated pH value to the user.

3 Technical Evaluation

To demonstrate practical use, we tested pHigment with real-world fluids—including food, drinks, and cleaning products—applying samples and measuring pH using a pH meter (Figure 4.a). Consumables were selected for being commonly ingested and more acidic than the recommended pH range for ingestion (6.5–8.5 [2]), while cleaning products were chosen for being highly basic, exceeding the recommended skin-safe pH range (4.0–5.0 [4]). The material exhibited clear color transitions depending on pH, and retained the change once dried (Figure 1.a).

We evaluated pHigment by applying a range of pH solutions that are in safe levels for human contact (4.5–9, in 0.5 increments) to material swatches and observing the resulting color changes (Figure 4.b). The solutions were prepared using H₂K₄O₇P for acidic (pH 4.3) and HK₂O₄P for alkaline (pH 9.7) conditions. Each was applied thinly with a soft nail sponge, similar to tools used in cosmetic applications. The app estimated values within 0.5–1.0 pH units of the ground truth, confirming its reliability.

To assess end-of-life sustainability, pHigment samples were placed in a home composting setup. Visible degradation, including edge breakdown and softening, began within two weeks (Figure 3), consistent with prior work on alginate-based biodegradability [3].

4 Discussion and Future Works

4.1 Potential Applications for Biosensing

pHigment demonstrates the potential of embedding biosensors into disposable cosmetics. pH is a key indicator in various health-related

¹pHigment Github: <https://github.com/InteractiveOrganismsLab/pHigment>

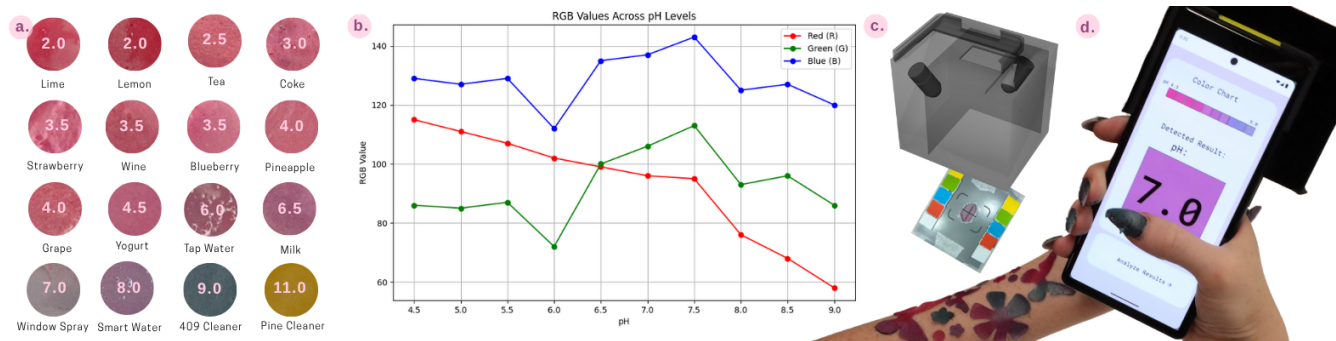


Figure 4: Testing pHigment: (a) exposure to real-world substances yields distinct color changes; (b) RGB values from samples plotted across pH range; (c) 3D enclosure, LEDs and color calibration card; (d) Readout from fingernail with app and light box.

conditions in bodily and external fluid-interactions. For instance, tattoo stickers placed on areas like the forehead, neck or underarm could detect changes in sweat pH, which may be influenced by hydration, physical exertion, or stress [14]. Eyeliner stickers, worn near the eyes, respond to tears and sweat enabling eye irritation to be monitored [13]. Nail stickers allow users to dip the sensor into fluids—such as saliva, beverages, cleaning products, or vaginal discharge—which is relevant to conditions like dental erosion, bacterial vaginosis, or yeast infections [6]. In occupational and environmental contexts, pH sensing can help identify harmful exposures—such as chemical residues in farming [7], acidic substances in food processing [29], or irritants in industrial work environments [1]. The ability to detect localized chemical changes on fluids that interact with the body through familiar cosmetic forms introduces a discreet and wearable format for situational biosensing.

4.2 DIY Process and Fabrication

pHigment draws from DIY biomaterials and digital fabrication practices to support replicability in small-scale settings. The fabrication uses widely available, food-safe ingredients—sodium alginate, red cabbage powder, glycerin—and accessible tools such as blenders, vinyl cutters, and cooking sheets. Importantly, the process does not involve any hazardous chemicals, making it safe for home, educational, and community settings. This approach is informed by the broader maker community’s work on alginate films and biopolymer experimentation [3]. By grounding the design in familiar DIY practices and non-toxic materials, the project supports at-home or community-based experimentation and fabrication, particularly relevant for education, participatory research, or low-resource environments. Additionally, designing new form factors can be easily done by cutting into custom single-use shapes such as full-face masks, under-eye patches, crease stickers, acne dots, jawline strips, nail wraps, vaginal moisturizer applicators, and panty liners.

4.3 Sustainable Biocosmetic Interfaces

Sustainability in HCI has often focused on extending the lifespan of electronics through biodegradable substrates that can be detached, often emphasizing environmental impact while underrepresenting social and economic dimensions [17]. The formulation relies on easy-accessible materials such as sodium alginate and red cabbage

powder—both that can be compost under natural conditions. Titanium dioxide is not biodegradable [5], however, our compost tests showed visible degradation within weeks, supporting short-use biocosmetics with reduced environmental persistence. Future work will explore alternatives materials for an opacifier.

Biocosmetic interfaces [18, 19] combine everyday cosmetics with biosensing to support discreet, on-body health monitoring. Prior work in this area has explored pH-responsive lipsticks[18], dental inserts[23], and permanent tattoos[28] that visualize chemical changes in bodily fluids. pHigment builds on this foundation to include biodegradable materials in the biosensor material formulation. This integration expands the design space for fluid-based interaction, showing how ephemeral, single-use cosmetics can serve both as health tools and as environmentally responsible materials.

4.4 Limitations

We plan to expand biomaterials to include chitosan, gelatin, and cellulose derivatives to compare structural integrity, biocompatibility, and compostability. Although the biosensing layer of pHigment is compostable, the double-sided sticker used for adhesion is not biodegradable and limits direct contact between sweat and the sensor, which may affect accuracy. Additionally, titanium dioxide used as an opacifier is not biodegradable, requiring alternative materials for improved sustainability.

Because biosensing data is shown via color changes, it may reveal personal health information and raise privacy concerns. However, prior studies [18] suggest users may not find this intrusive. Future studies can explore privacy and social acceptance further. While pHigment shows promise as a cosmetic biosensor, its app requires an enclosure and supports a limited pH range. Upcoming work will improve the color-analysis software for more consistent results.

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